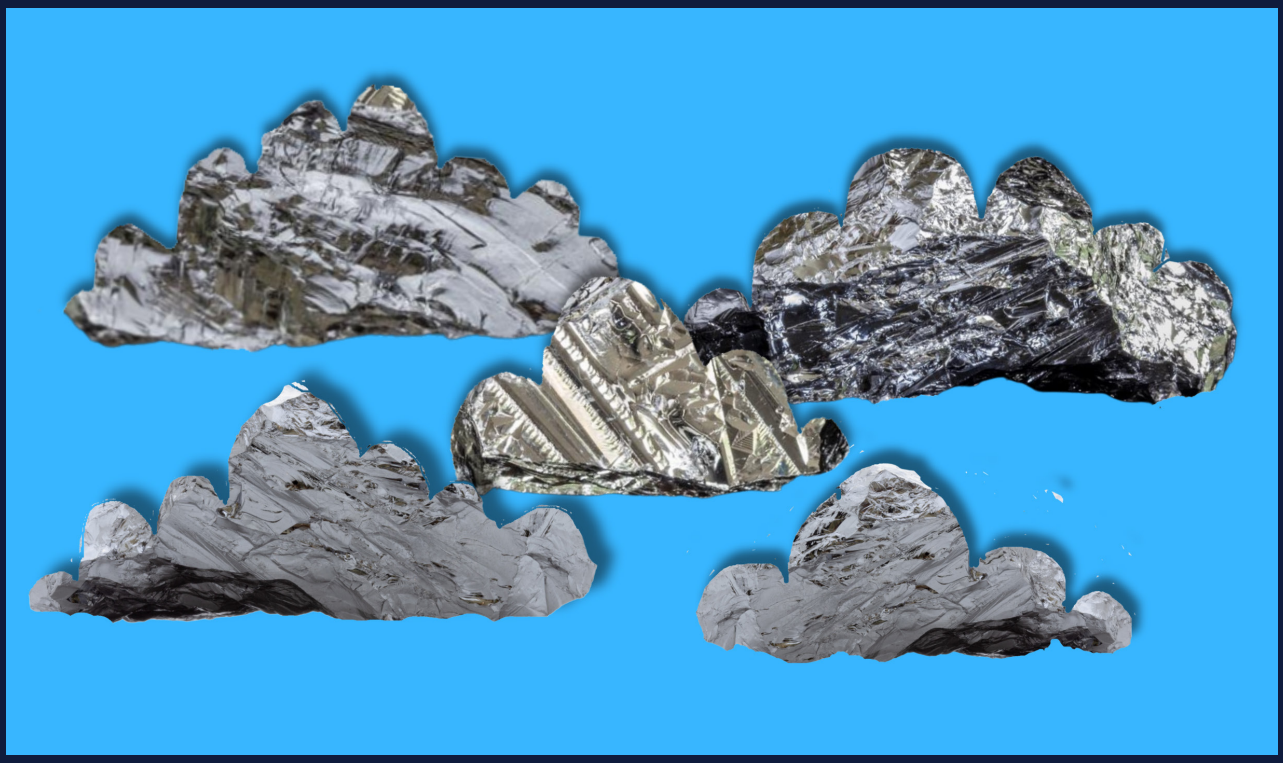
# Cloud computing and the DMA

## A path to effective enforcement

Submission to the European Commission

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## Summary

In 2025, the European Commission took the critical step of initiating three market investigations into cloud computing services under the Digital Markets Act (DMA). Two have yielded preliminary opinions that Microsoft's and Amazon's cloud computing businesses should be designated as core platform services. The third, assessing whether current obligations under the DMA are effective in addressing practices that limit contestability or fairness in the cloud sector, is the focus of this submission.

Europe relies on three non-European tech giants for over 70% of its cloud services. Its businesses and governments are paying excessively for this concentrated infrastructure while running the risk of being spied on, manipulated or directly attacked by foreign corporations and governments. Moreover, the so-called cloud "hyperscalers" – Amazon, Google and Microsoft – engage in a wide range of anticompetitive practices to ensure they remain on top, preventing competitors (including European ones) from scaling. Without structural change in the cloud sector, AI will only amplify these risks.

The need to address today's highly concentrated, U.S. dominated cloud market grows more urgent by the day. Geopolitical tensions continue to rise, highlighting the dangers of Europe's tech dependencies and increasing the need to build decentralised and sovereign digital infrastructure. At the same time, the AI boom has triggered an explosion in data centre construction: $300bn is expected to be invested in European data centre construction to keep pace with AI's demands. As the public sector and industry accelerate their cloud adoption, it is essential that they do so in a fair and contestable market, not a highly concentrated one. The DMA has a critical role to play in ensuring that this market takes shape.

At the time of the DMA's drafting, legislators were primarily focused on multi-sided consumer platforms including social media, app stores, operating systems, and search engines. This is reflected in many of the Act's obligations which, in their current form, are either unsuited or unable to address many of the most problematic forms of anticompetitive conduct in the cloud sector.

**For these reasons we urge the following amendments to the DMA:**

- Expand Article 5(2)'s prohibition on the processing, combination and cross-use of data to non-personal data.
- Strengthen Article 5(8) by extending its scope to cover economically equivalent forms of bundling and by removing its restriction to CPS-to-CPS tying.
- Extend Article 6(5) beyond ranking-related conduct to cover functionally equivalent forms of self-preferencing (pricing, contractual, and technical).
- Extend Article 6(7) to allow the imposition of interoperability requirements for cloud services.
- Extend the FRAND obligation in Article 6(12) to cover cloud services.

## Anatomy of a cloud gatekeeper

The gatekeeper power of cloud hyperscalers operates at multiple levels.

First, they own, lease, and run physical infrastructure (data centres, undersea cables, network hardware) at a scale that cannot be replicated by challengers. Google, one of the most profitable companies in the history of the world, was only able to catch up with Amazon and Microsoft by cross subsidising its cloud division for fifteen years. For businesses that need global reach, low latency, and guaranteed uptime, there is often no credible outside option.

On top of this infrastructural dominance, hyperscalers provide the digital resources necessary to run software (compute, storage, and networking) using their own proprietary implementations. While these services are largely functionally equivalent, they have distinct application programming interfaces (APIs) and configuration formats. This means that a business that has built its infrastructure around AWS's APIs faces significant engineering costs to migrate to Azure's equivalents not because the underlying concepts differ, but because the interfaces and data formats do not translate. In other words, even if a company builds its own software, it easily finds itself technically – and economically – locked into a single provider's ecosystem.

This lock-in only becomes more extreme as one moves up the stack. Alongside basic compute, storage and networking resources, hyperscalers also offer managed services: managed databases, message queues, content delivery networks, identity and access management (IAM), monitoring, logging, security tooling, and more.

Most businesses today use some combination of these managed services: it is simply more convenient and – at least in the short term – cheaper than self-managing open-source alternatives. In doing so, they weave the hyperscalers' proprietary services into the core of their operations. The more managed services a business adopts, the more tightly it is woven into a single provider's ecosystem, and the more it is dependent on that provider to run its software, and therefore its business. Not only this, but the developer toolkits and methods used to create systems differ from provider to provider, reinforcing lock-in as teams are built around the provider they use.

Beyond the closed ecosystems they create and control, hyperscalers also dominate the enterprise software economy. Just as the Apple and Google app stores do for consumer software, the hyperscalers – via AWS Marketplace, Azure Marketplace, and Google Cloud Marketplace – control the dominant channels through which enterprise software vendors reach customers.

Enterprise buyers often commit substantial spending to hyperscalers via multi-year contracts (AWS Enterprise Discount Programme, Azure Consumption Commitments, and Google Consumption Commitments) which they can then allocate to purchases of third-party software services on (and only on) the hyperscaler's marketplace, reinforcing lock-in. In this way, hyperscalers become gatekeepers not only of the infrastructure that the enterprise software economy runs on, but also of the main distribution channels for that software.

## Cloud and AI: why we need to get this right, right now

AI is fundamentally a cloud-based technology. While smaller AI models can run locally, generative AI's computational demands mean that the vast majority of workloads will be carried out on centralised cloud infrastructure. In this sense the AI boom is, more than anything else, a cloud boom: a $7 trillion one if McKinsey's projections are correct.

The relationship runs in both directions. The growth of AI is turbocharging demand for cloud services, but the oligopolistic structure of the cloud sector will equally shape how AI develops and who captures its value.

AI does not sit on top of cloud infrastructure as a separate layer: it is built into it. The dominant AI deployment platforms for enterprise are Microsoft Azure, AWS Bedrock, and Google Cloud Platform. The proprietary GPU clusters required to train and run large models at scale exist, at the necessary scale, only within hyperscaler infrastructure.

A startup or enterprise building AI-powered products is therefore dependent on hyperscaler infrastructure not as a matter of convenience but of necessity, and that dependency is subject to all the same lock-in dynamics described above in the context of traditional enterprise software. Any European AI company seeking to distribute AI models or AI services globally currently has no choice but to go through the hyperscalers. Once they are locked into a hyperscaler's ecosystem they will find it very difficult to escape, further entrenching this dominance.

With frontier models from leading AI labs, such as OpenAI and Anthropic, currently distributed via the hyperscalers they depend on for investment and compute, rapid uptake of frontier AI is further entrenching the hyperscalers' market dominance – and further deepening Europe's dependencies.

Agentic AI is likely to make things worse. As AI systems increasingly operate autonomously across business workflows, they do so through integrations with enterprise software stacks that are themselves cloud hosted. The connective tissue of agentic AI, if developed by the hyperscalers, will naturally reinforce lock-in and closed ecosystems.

Making the DMA fit for the cloud is therefore not only about the present moment, but also about the future of AI in Europe.

## How to effectively apply the DMA to cloud services

Anti-competitive practices in the cloud sector have been extensively documented by the Open Markets Institute, national regulators, and cloud competitors.

These include:

- Discriminatory pricing and contractual terms;
- Egress fees;
- Unnecessary technical barriers to switching;
- Tying and bundling;
- Self-preferencing;
- Anti-competitive cloud credit schemes and committed spend discounts;
- Copycatting of customer services.

## Cloud-ready DMA provisions

Several existing DMA obligations can already address harmful practices by dominant cloud providers.

***Article 5(4): anti-steering***

Under the anti-steering obligation, designated cloud providers must allow business users (both contractually and technically) to communicate and promote offers and sign contracts, including under different conditions, to end users they acquire via its core platform service or through other channels. In the cloud context, this would ensure that hyperscalers cannot stop enterprise developers from directing their own end users to purchase, renew, or transact directly with them – at different prices or terms – outside of the gatekeeper's cloud marketplace.

***Article 6(2): data use***

Designated cloud providers would be prevented from using confidential commercial data generated by their business users (such as software developers) to benefit their cloud and non-cloud services, including by creating copycat software.

***Article 6(6): switching***

Designated providers would be barred from technically or contractually blocking customers from switching to, or running, competing software/services on their cloud infrastructure (e.g., discouraging multi-cloud tooling, blocking third-party services from functioning effectively, egress fees, unfair retention via excessive cloud credits).

***Article 6(9): data portability***

Designated gatekeepers will need to ensure the full portability of business and end user data to competitors free of charge. *While this provision is already legally applicable to the cloud sector, the key will be in how it is applied.*

Portability is only meaningful if the metadata (which defines the structure, format, and relationships of the underlying data) is exported alongside the data itself in a flat, interoperable, readable format.

Data that is technically transferable but locked into proprietary metadata structures isn't in fact usable by competitors.

The DMA gives the Commission power to order the development of technical standards (Article 48), which can then be mandated via an implementing act (Article 8). The Data Act – potentially joined by CADA – also encourages standard-setting in the cloud sector. Together, these instruments must be used to create and enforce a single, open, flat-format metadata export standard applied to the wider cloud sector and enforced against gatekeepers.

***Article 13: anti-circumvention***

The powers conferred under the DMA's anti-circumvention provision can help address hyperscaler conduct designed to offset the impact of pro-competitive remedies.

Cloud gatekeepers are likely to engage in a range of circumventing behaviours to avoid making their cloud services genuinely portable and interoperable. For example, hyperscalers that use cloud credits to prevent customers from using portability and interoperability provisions to switch provider, could be caught by Article 13.

## Provisions requiring amendment to effectively tackle anticompetitive practices

Several DMA provisions that could apply to cloud gatekeepers will not do so in their current form. This is fully or partially the case for:

***Article 5(2): prohibition on processing, combining, and cross-using data across services***

Article 5(2) is designed to address gatekeepers' structural advantage in accumulating and leveraging data across ecosystems. However, its current limitation to personal data significantly narrows its effectiveness in the cloud context.

Cloud services are predominantly B2B in nature, with some of the most commercially valuable datasets being non-personal, such as operational, transactional, or usage data generated by business users. Large-scale access to such business data can be used to optimise infrastructure, refine services, and develop adjacent offerings. Article 5(2) thus fails to address the ability of gatekeepers to aggregate and cross-use such non-personal business data to strengthen their services.

This limitation risks becoming even more problematic considering the growing importance of AI. Cloud providers are uniquely positioned to leverage vast amounts of non-personal data to train and improve AI models, reinforcing feedback loops between data access and service quality. In its current form, Article 5(2) fails to capture an increasingly central mechanism through which competitive advantages are entrenched within the broader cloud and AI stack.

**Article 5(2)'s prohibition on the processing, combination and cross-use of data should be expanded to include non-personal data.**

***Article 5(8): prohibition on bundling***

Article 5(8) prohibits gatekeepers from requiring users to subscribe to or register with additional core platform services as a condition for accessing another CPS. While this targets explicit forms of tying within gatekeeper ecosystems, its scope is relatively narrow and is unlikely to tackle the bundling seen in the cloud sector.

First, the provision focuses on formal subscription or registration requirements. In doing so it fails to capture de facto bundling achieved through pricing strategies or technical integrations that make standalone use commercially unattractive.

Second, it applies only to the bundling of CPSs with other CPSs, excluding situations where gatekeepers combine CPSs with non-CPS services or functionalities.

These limitations are particularly relevant in the cloud sector, where providers commonly bundle infrastructure, platform, and software services to create integrated ecosystems and induce lock-in. Such practices often do not rely on formal subscription to multiple CPSs but can nonetheless generate significant switching costs and foreclosure effects.

**Article 5(8) be expanded to cover economically equivalent forms of bundling and by removing the limitation to CPS-to-CPS tying.**

***Article 6(5): self-preferencing***

Article 6(5) prohibits gatekeepers from favourably treating their own services in ranking, indexing, and crawling. The provision does not extend to other forms of self-preferencing, such as pricing strategies, contractual terms, or technical integration that give exclusive advantages to a gatekeeper's own services.

This limitation is particularly relevant in the cloud sector, where hyperscalers can reinforce their market position through practices such as:

- discriminatory licensing conditions, for example where software licensing is more costly or restrictive when deployed on rival cloud infrastructure (a tactic used by Microsoft)
- preferential technical integration and commercial advantages for proprietary in-house services, including earlier integration for first-party tools and default configurations favouring the provider's own offerings;

**Article 6(5) should be expanded beyond ranking-related conduct to cover functionally equivalent forms of self-preferencing across pricing, contractual, and technical dimensions.**

***Article 6(7): interoperability***

Article 6(7) requires gatekeepers to provide business users and alternative service providers access to, and interoperability with, certain hardware and software functionalities of operating systems and virtual assistants. Its scope is limited to those CPSs and thus cannot currently be applied to cloud computing services.

In cloud services, interoperability issues arise across infrastructure, platform, and software layers, including data portability constraints, proprietary APIs, incompatible managed services and technical barriers to multi-cloud deployment. Article 6(7) is designed to ensure that third-party services can access functionalities within a gatekeeper's ecosystem (intra-ecosystem interoperability) rather than to require interoperability between competing ecosystems (inter-ecosystem interoperability). As a result, it does not address many of the technical barriers that reinforce cloud lock-in.

These barriers increase switching costs and enable gatekeepers to favour their own cloud software and platform services. They thus impede the growth of competitors offering interoperable or multi-cloud software solutions as well as competitors offering cloud infrastructure services.

**Article 6(7) should be extended to cloud computing, or equivalent interoperability rules for cloud computing services should be introduced. This should include requirements on interoperable APIs, compatibility with open standards, and access to technical interfaces needed to support data portability and multi-cloud deployment.**

***Article 6(12): FRAND access for business users***

Article 6(12) requires gatekeepers to apply fair, reasonable, and non-discriminatory (FRAND) general conditions of access for business users to CPSs. Currently however, it only applies to software application stores, online search engines and online social networking services.

Extending Article 6(12) to cloud gatekeepers would, for instance, help ensure that they cannot give their own services preferential technical access or product placement on their cloud platforms.

In relation to AI, a FRAND regime would help secure fair access to compute resources for all market participants, preventing gatekeepers from granting preferential, exclusive, or conditional access to the high-performance infrastructure essential for training and deploying AI models, whether to their AI services or those of favoured partners.

FRAND obligations could also be used to constrain the use of committed spend discounts and cloud credit schemes by requiring that such pricing schemes are transparent, objectively justified, and applied on non-discriminatory terms. While not prohibiting volume-based discounts as such, this would help prevent their abuse. In the cloud context, this could help prevent large-scale discounts from distorting competition between providers or foreclosing smaller rivals.

**Article 6(12) should be extended to cloud services to ensure that access to critical cloud infrastructure and platforms is provided on FRAND terms and does not reinforce lock-in or distort fair competition.**

## The role of a strong DMA in the wider cloud regulatory context

With the Data Act entering into application in 2025, the EU has two regulatory regimes that seek to address unfair practices in the cloud sector. Our proposed amendments would create some overlap between the Data Act and the DMA, particularly with respect to interoperability and data portability. Due to the fundamentally asymmetric design of the DMA, however, this should not be seen as a problem.

While the Data Act applies horizontally across the cloud sector, the DMA only imposes obligations on designated gatekeepers. This asymmetry enables a more targeted regulatory approach, focused on systemically significant actors whose conduct shapes (and distorts) market outcomes. In practice, it allows for tailored obligations to be imposed on hyperscalers – such as enhanced interoperability, restrictions on self-preferencing, or FRAND access conditions – without imposing comparable compliance burdens on smaller providers.

The DMA also offers advantages in terms of enforcement. Unlike the Data Act's implementation by designated national competent authorities, the DMA's centralised enforcement reduces the risk of divergent interpretation and enforcement across Member States. This is particularly significant in technically complex areas such as cloud switching and interoperability. It also allows Europe to respond to its cloud dependency problems with greater speed and coherence.

Early implementation of the Data Act already points to uneven progress. While some Member States, such as France, have moved relatively quickly to designate authorities and operationalise key provisions, others remain significantly further behind in the process.

Overlap can thus create powerful complementarities between the two regimes. Centralised, targeted enforcement of obligations on hyperscalers by the Commission under the DMA can complement and provide impetus to Member State enforcement of the Data Act for other actors, indirectly promoting harmonisation. Aligning the DMA and the Data Act's obligations should therefore become a priority.

The DMA also goes well beyond the Data Act, which does not address a range of sector-specific anti-competitive practices deployed by cloud gatekeepers including self-preferencing, tying and bundling, and discriminatory treatment of customers.

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